Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

Department of CSE

B.Tech. CSE Trimester -5

2520030054
PERUGU MANISHA

Cloud Infrastructure and Services
25CS2206E

LAB RECORD

2026

EXPERIMENT – 2

Title:
a) Enabling Multi-Factor Authentication (MFA) to Secure Your Access
b) Creating First Linux Instance

c) S3 Bucket Creation and Upload Objects
I. Create an S3 Bucket for file storage.
II. Upload 5 objects with different file extensions.

2(a) Enabling Multi-Factor Authentication to Secure Your Access

Aim:

To enable Multi-Factor Authentication (MFA) for an AWS account to provide an additional layer of security and protect the account from unauthorized access.

Description:

Multi-Factor Authentication (MFA) is a security feature that requires users to provide two or more verification factors to access an account. In AWS, MFA adds an extra layer of protection along with the username and password.

It helps prevent unauthorized access even if the password is compromised. AWS supports virtual MFA applications such as Google Authenticator and Microsoft Authenticator.

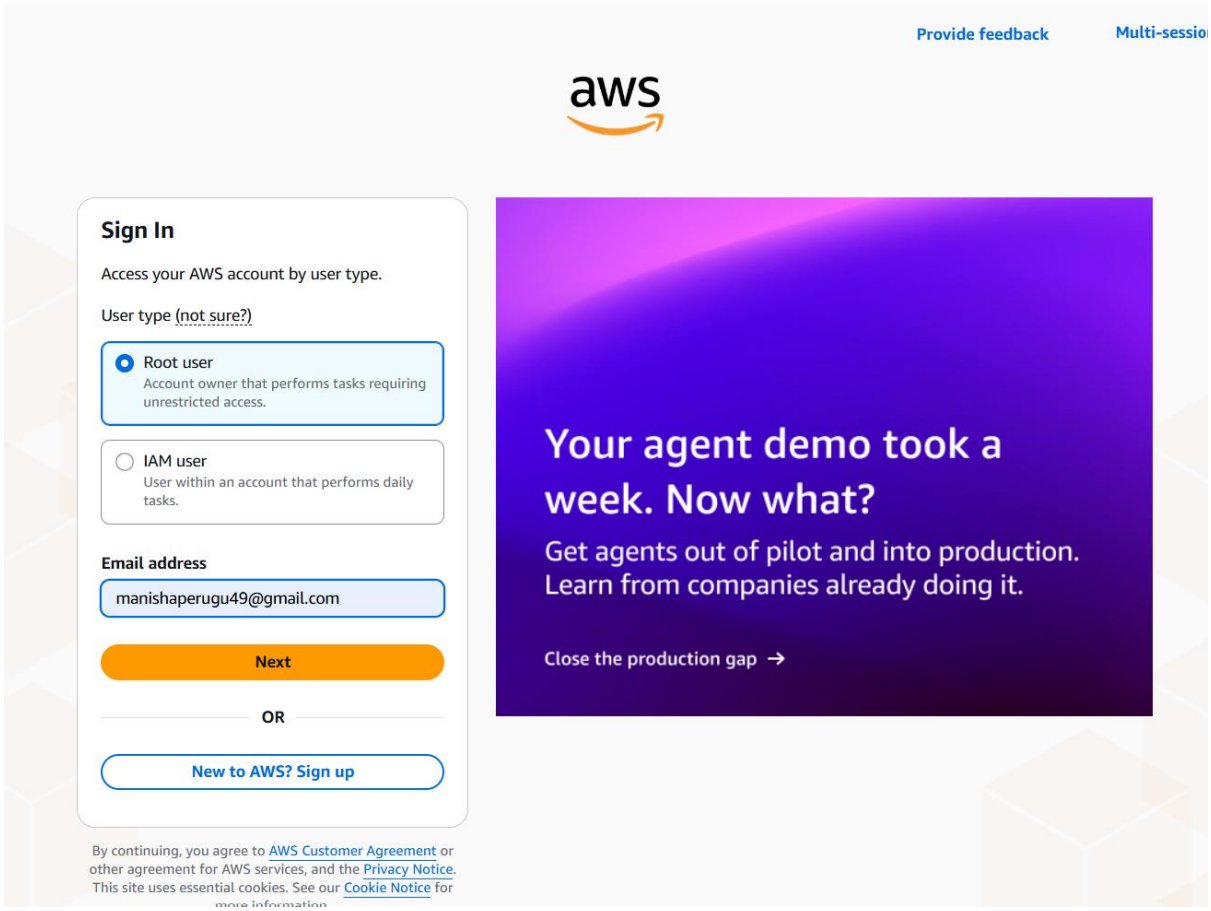
Steps:

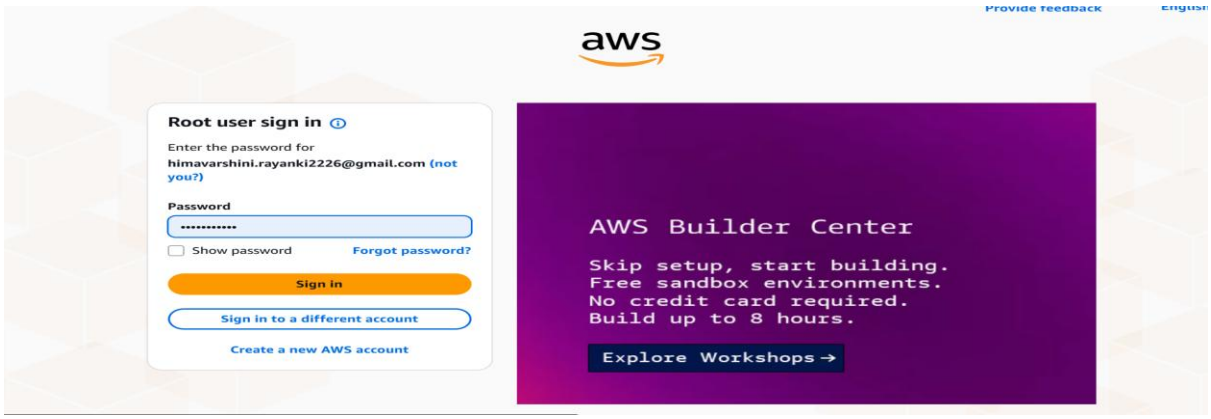
1. Open the AWS Management Console using a web browser.
2. Log in to the AWS account using your credentials.
3. Search for IAM (Identity and Access Management) in the AWS search bar.
4. Open the IAM dashboard.
5. Select Users from the left-side navigation panel.
6. Click on your IAM username.
7. Open the Security credentials tab.
8. Find the Multi-Factor Authentication (MFA) section.
9. Click Assign MFA device.
10. Enter a device name and select Authenticator app.
11. Click Next to display the QR code.

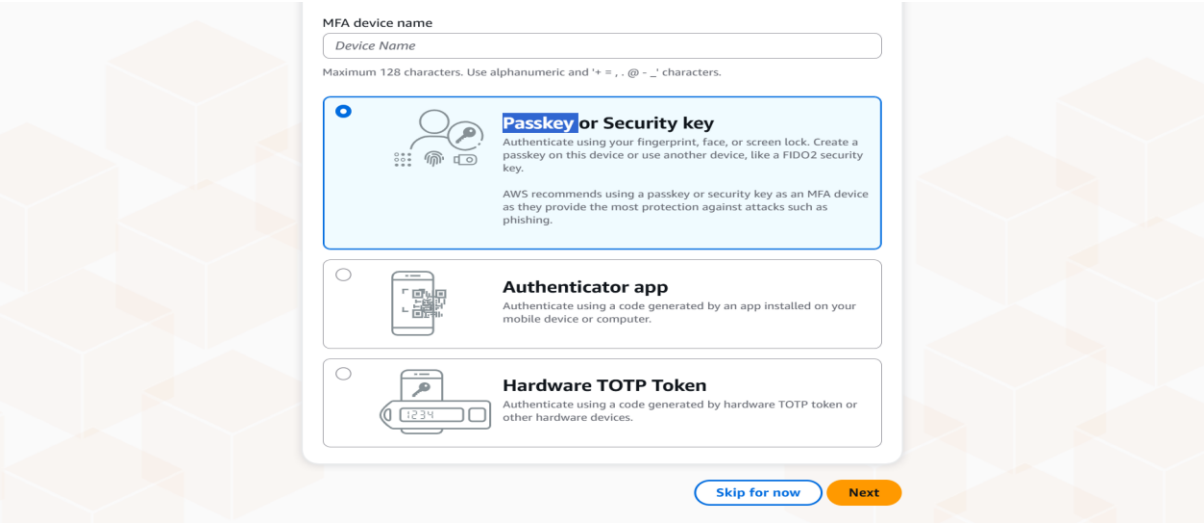
12. Open Google Authenticator or Microsoft Authenticator on your mobile phone.
13. Scan the QR code displayed on the AWS console.
14. Enter two consecutive MFA codes generated by the authenticator application.
15. Click Add MFA to complete the configuration.
16. Verify that the MFA device is successfully assigned.

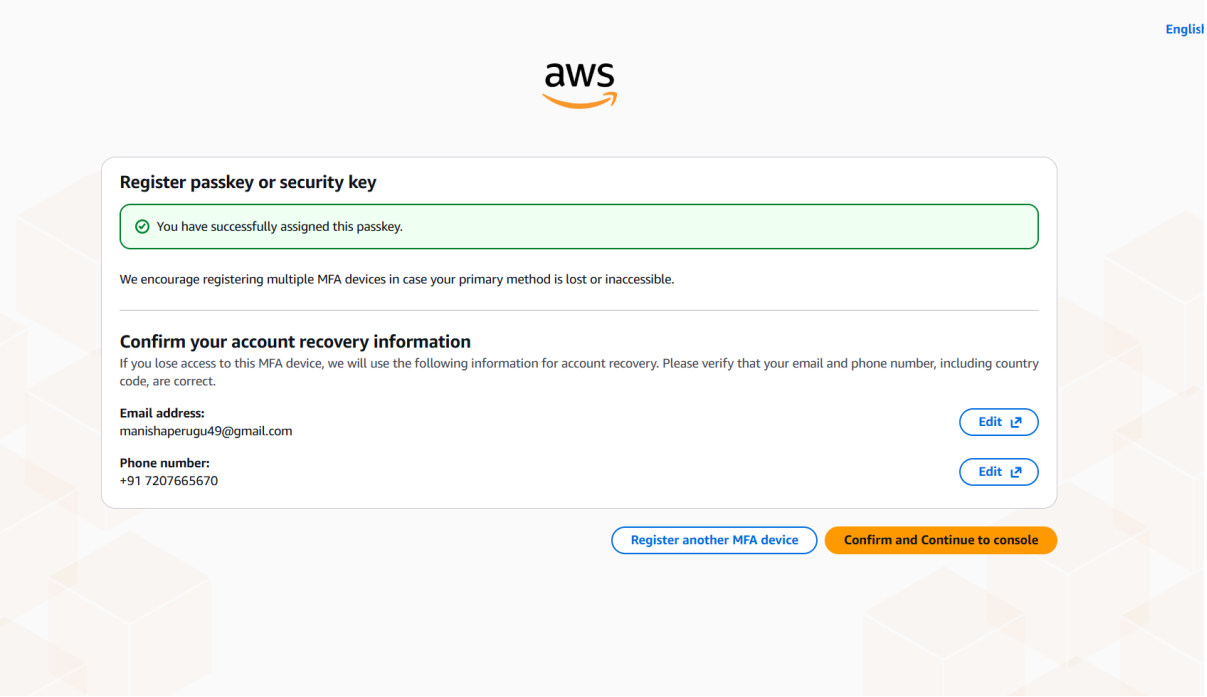
Output:

Multi-Factor Authentication was successfully enabled for the AWS account, providing an additional layer of security against unauthorized access.


Provide feedback
Multi-sessio
aws
Sign In
Access your AWS account by user type.
User type (not sure?)
Root user
Account owner that performs tasks requiring unrestricted access.
IAM user
User within an account that performs daily tasks.
Email address
manishaperugu49@gmail.com
Next
OR
New to AWS? Sign up
By continuing, you agree to AWS Customer Agreement or other agreement for AWS services, and the Privacy Notice. This site uses essential cookies. See our Cookie Notice for
Your agent demo took a week. Now what?
Get agents out of pilot and into production. Learn from companies already doing it.
Close the production gap →


aws
Root user sign in
Enter the password for himavarshini.rayanki2226@gmail.com (not you?)
Password
Show password
Forgot password?
Sign in
Sign in to a different account
Create a new AWS account
AWS Builder Center
Skip setup, start building.
Free sandbox environments.
No credit card required.
Build up to 8 hours.
Explore Workshops →





2(b) Creating First Linux Instance

Aim:

To create and launch the first Linux virtual machine instance using Amazon EC2 (Elastic Compute Cloud) in AWS.

Description:

Amazon EC2 is a cloud computing service provided by AWS that allows users to create and manage virtual servers called instances.

A Linux EC2 instance provides a Linux-based virtual environment where users can execute commands, install software, host applications, and perform server administration tasks.

Steps :

1. Log in to the AWS Management Console.
2. Search for EC2 in the AWS search bar and open the EC2 dashboard.
3. Select Instances from the left-side navigation panel.
4. Click on Launch instances.
5. Enter a suitable name for the instance, such as My-Linux-Instance.
6. Under Application and OS Images (AMI), select Amazon Linux.
7. Choose an eligible instance type, such as t2.micro or t3.micro, depending on availability and Free Tier eligibility.
8. Under Key Pair, click Create new key pair.
9. Enter a key pair name and select the .pem format.
10. Download and securely save the key pair file.
11. Configure the network settings and enable SSH access (port 22) only from your IP address.
12. Review the storage configuration.
13. Click Launch instance.
14. Wait until the instance state changes to Running.
15. Select the created instance and click Connect.
16. Choose the EC2 Instance Connect option if supported, or connect using SSH and the downloaded key pair.
17. After connecting, execute basic Linux commands such as:

pwd

ls

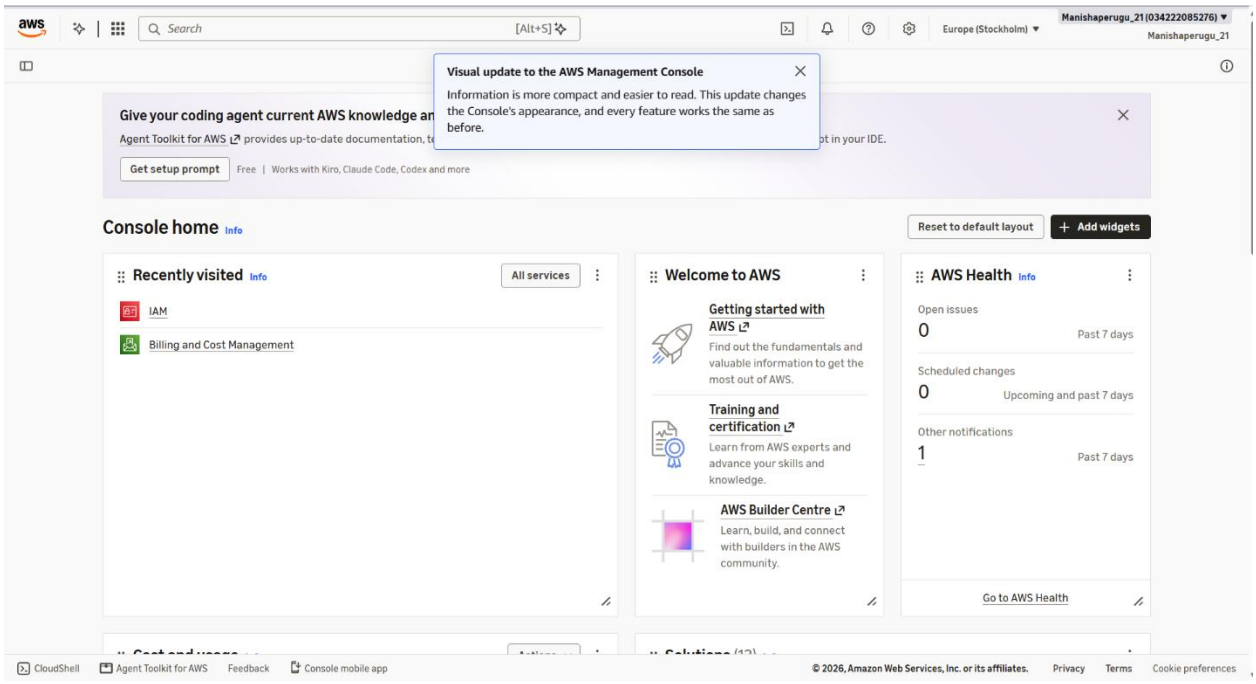
whoami

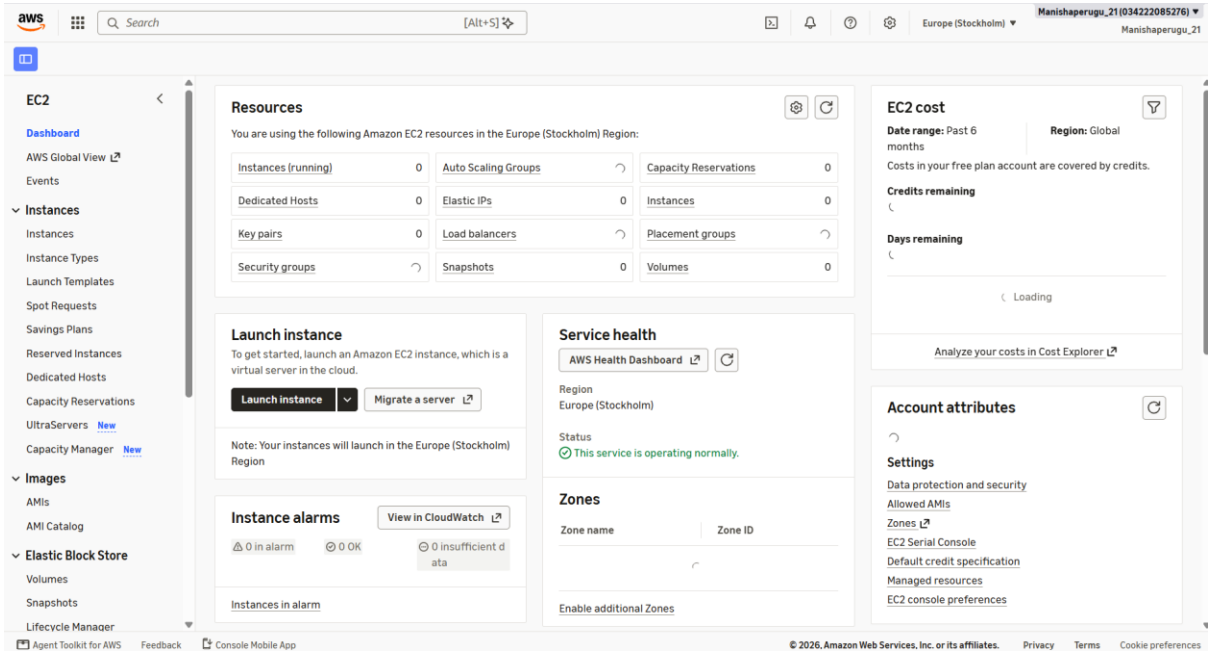
uname -a

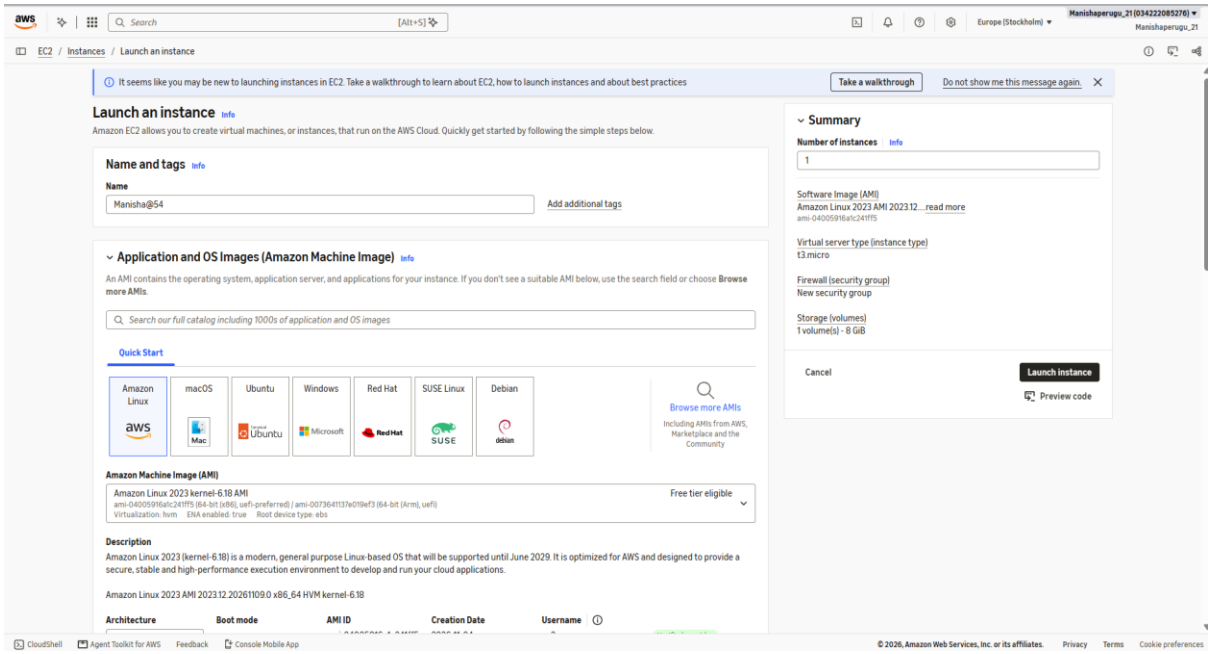
18. Verify that the Linux instance is running and responding to commands.

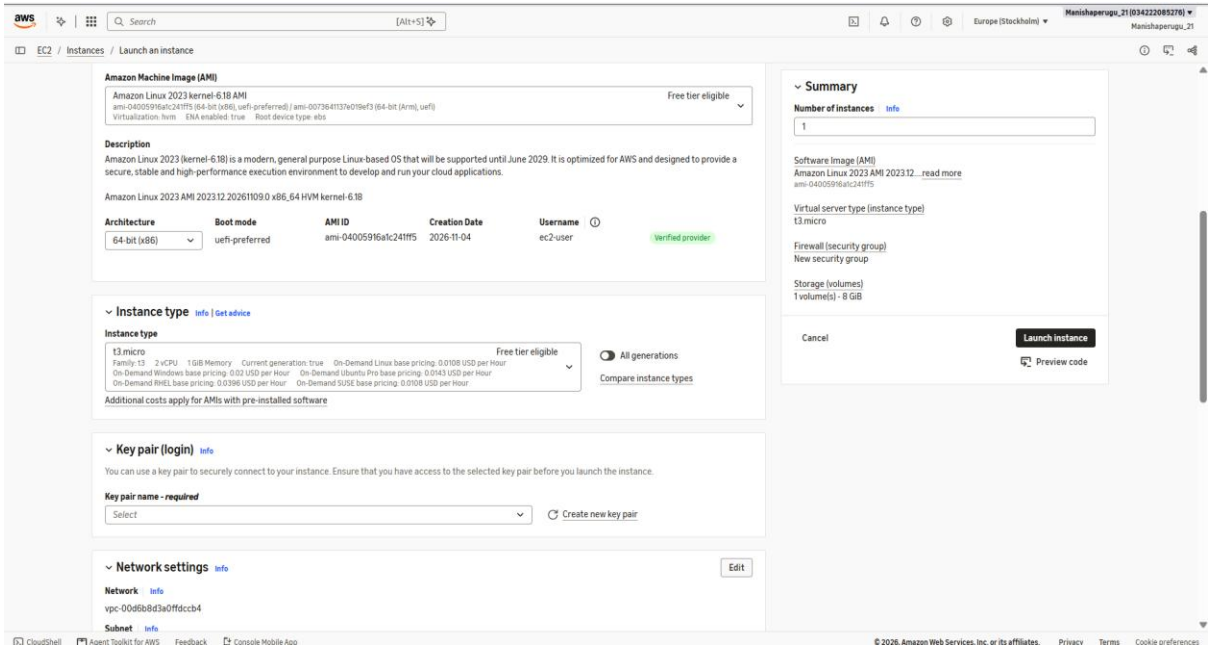
Output:

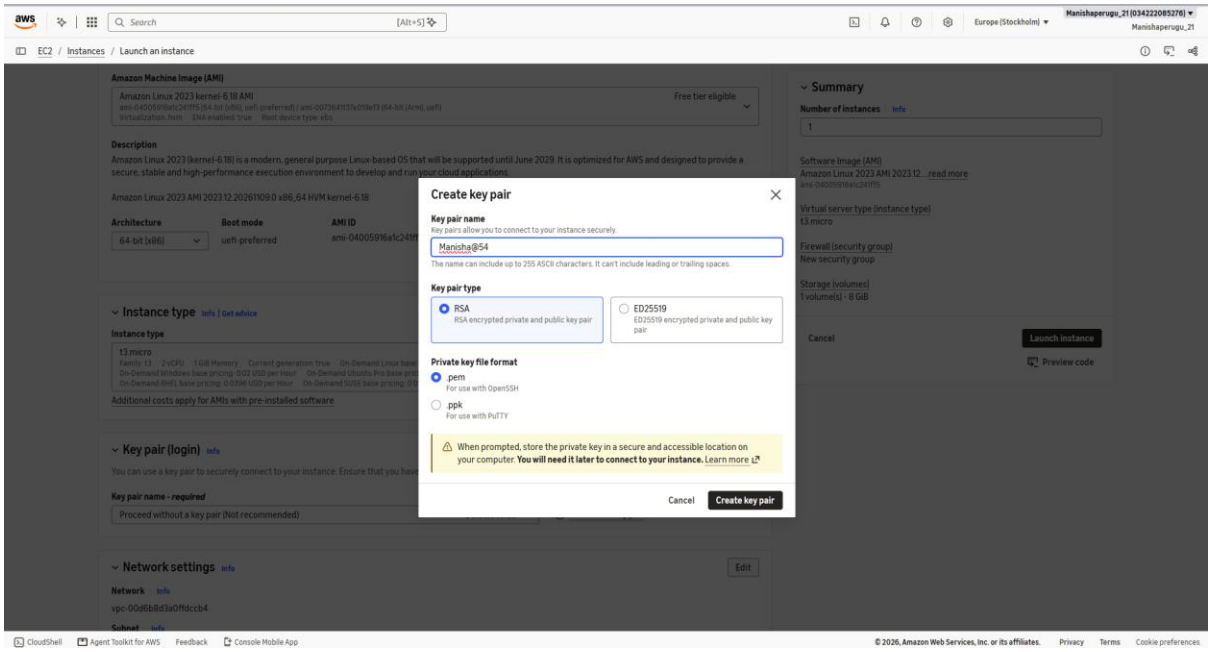
A Linux virtual machine instance was successfully created and launched using Amazon EC2 in AWS. The instance was accessed successfully, and basic Linux commands were executed.

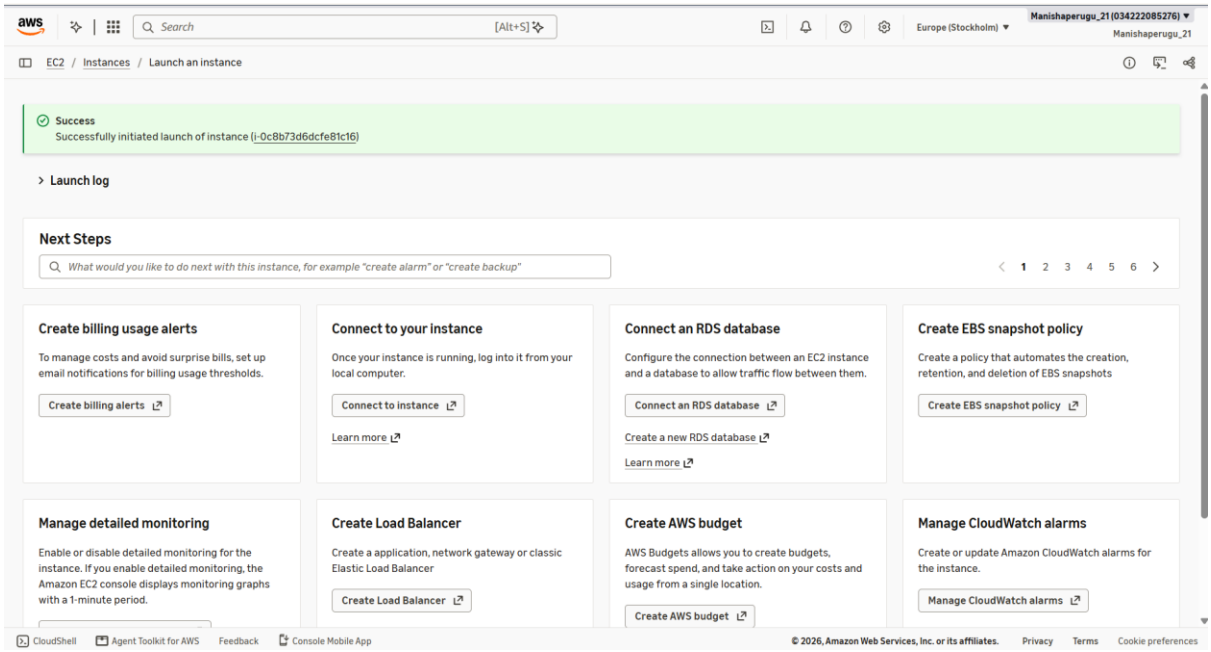


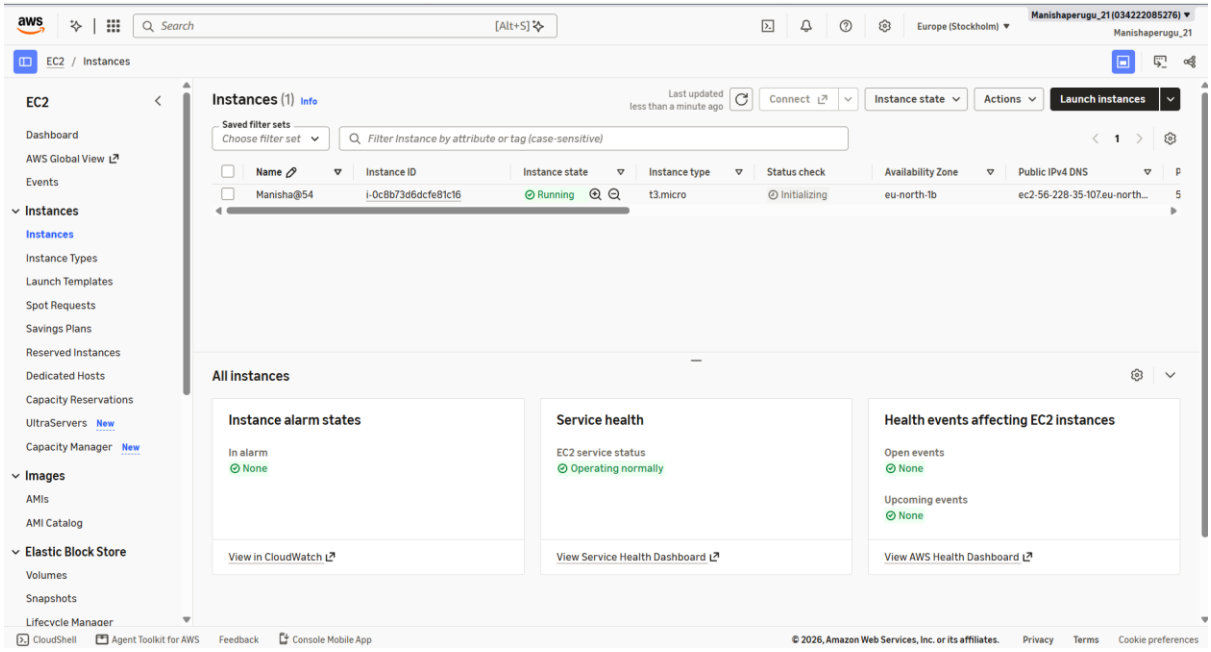


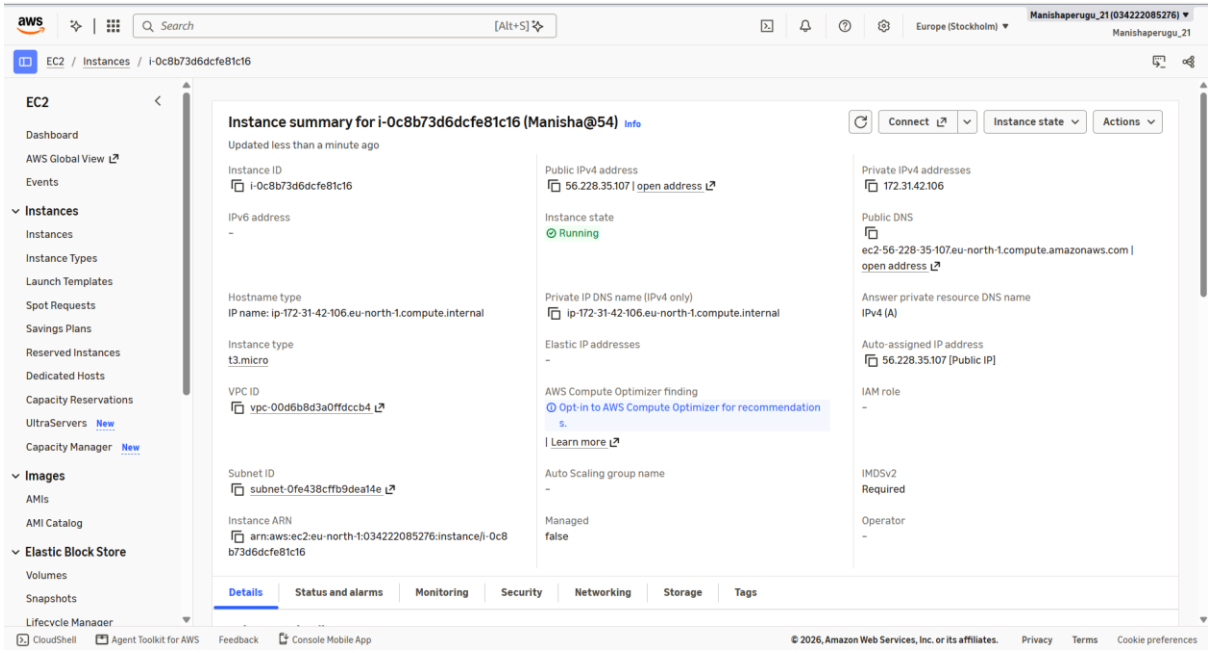

Launch an instance
Name and tags
Manisha@54
Application and OS Images (Amazon Machine Image)
Amazon Linux 2023 kernel-6.18 AMI
Free tier eligible
Summary
Number of instances
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12...read more
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code

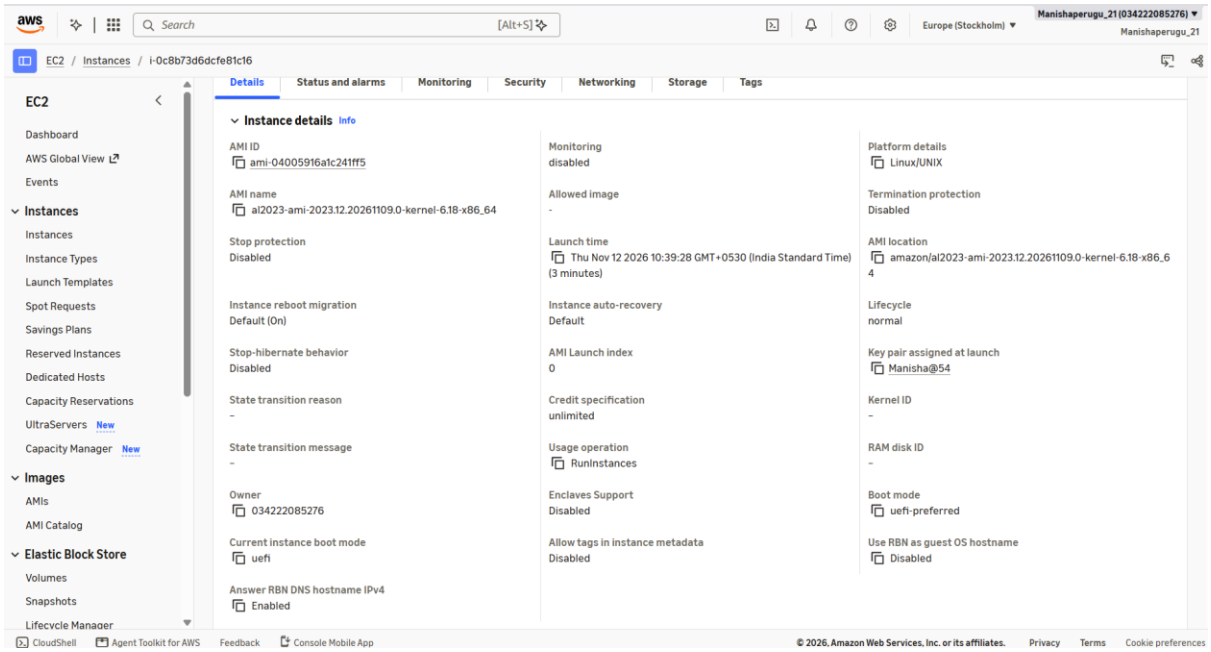

Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.18 AMI
Free tier eligible
Description
Amazon Linux 2023 (kernel-6.18) is a modern, general purpose Linux-based OS that will be supported until June 2029. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.
Amazon Linux 2023 AMI 2023.12.20261109.0 x86_64 HVM kernel-6.18
Architecture
64-bit (x86)
Boot mode
uefi-preferred
AMI ID
ami-04005916a1c241ff5
Creation Date
2026-11-04
Username
ec2-user
Verified provider
Instance type
t3.micro
Free tier eligible
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login)
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
Select
Create new key pair
Network settings
Edit
Network
vpc-00d6b8d3a0ffdccb4
Subnet
Summary
Number of instances
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12...read more
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code

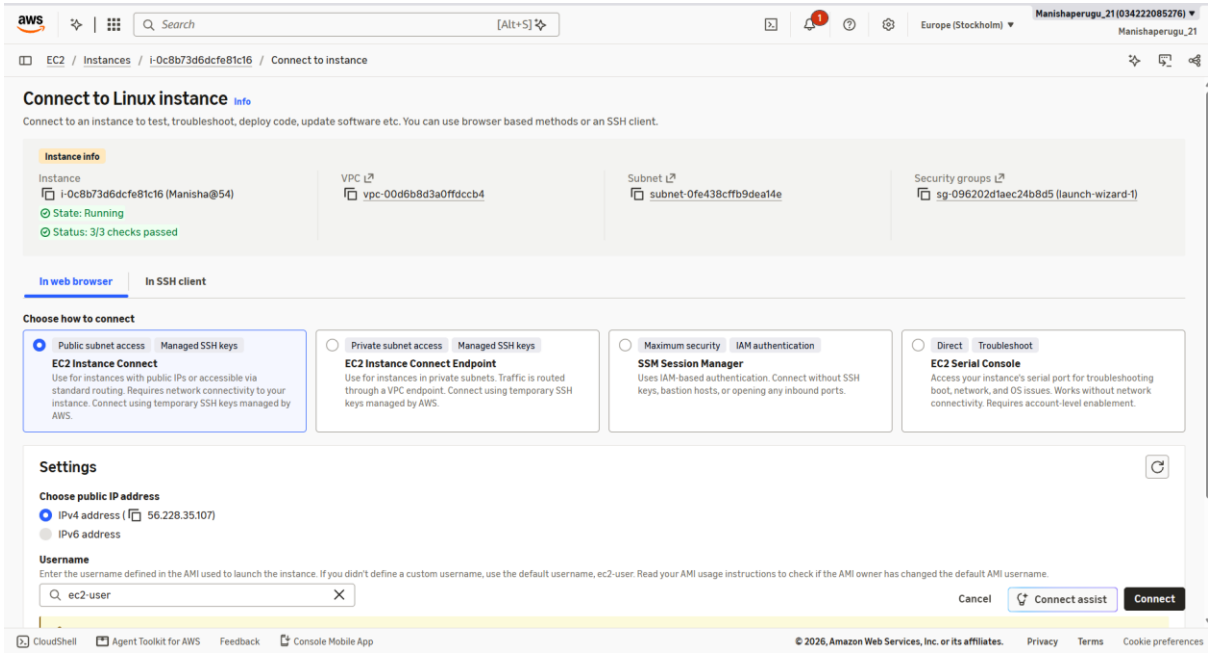

Create key pair
Key pair name
Manisha@54
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more
Cancel
Create key pair
Summary
Launch instance
Preview code

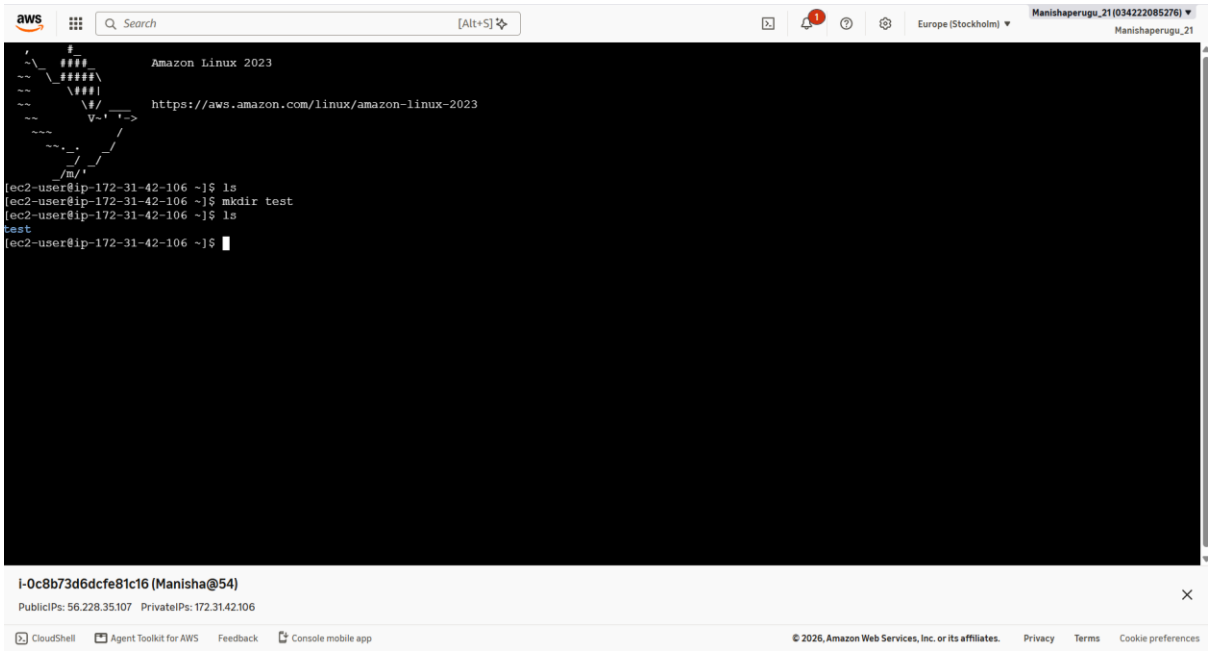

Success
Successfully initiated launch of instance (i-0c8b73d6dcfe81c16)
Launch log
Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"
Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more
Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy
Manage detailed monitoring
Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.
Create Load Balancer
Create a application, network gateway or classic Elastic Load Balancer
Create Load Balancer
Create AWS budget
AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.
Create AWS budget
Manage CloudWatch alarms
Create or update Amazon CloudWatch alarms for the instance.
Manage CloudWatch alarms

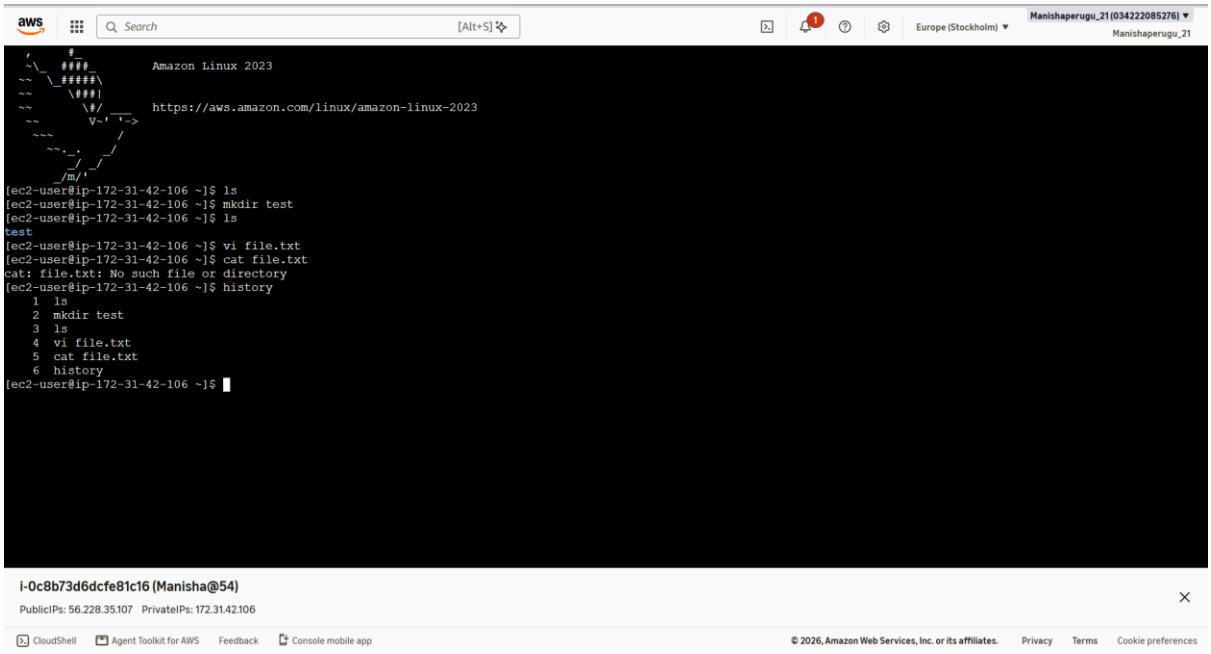

Search
[Alt+S]
Europe (Stockholm)
Manishaperugu_21 (034222085276)
Manishaperugu_21
EC2 / Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Instances (1) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
Manisha@54
i-0c8b73d6dcfe81c16
Running
t3.micro
Initializing
eu-north-1b
ec2-56-228-35-107.eu-north...
All instances
Instance alarm states
In alarm
None
View in CloudWatch
Service health
EC2 service status
Operating normally
View Service Health Dashboard
Health events affecting EC2 instances
Open events
None
Upcoming events
None
View AWS Health Dashboard
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search
[Alt+S]
Europe (Stockholm)
Manishaperugu_21 (034222085276)
Manishaperugu_21
EC2 / Instances / i-0c8b73d6dcfe81c16
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Instance summary for i-0c8b73d6dcfe81c16 (Manisha@54) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-0c8b73d6dcfe81c16
Public IPv4 address
56.228.35.107 | open address
Private IPv4 addresses
172.31.42.106
IPv6 address
-
Instance state
Running
Public DNS
ec2-56-228-35-107.eu-north-1.compute.amazonaws.com | open address
Hostname type
IP name: ip-172-31-42-106.eu-north-1.compute.internal
Private IP DNS name (IPv4 only)
ip-172-31-42-106.eu-north-1.compute.internal
Answer private resource DNS name
IPv4 (A)
Instance type
t3.micro
Elastic IP addresses
-
Auto-assigned IP address
56.228.35.107 [Public IP]
VPC ID
vpc-00d6b8d3a0ffdccb4
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
Learn more
IAM role
-
Subnet ID
subnet-0fe438cffb9dea14e
Auto Scaling group name
-
IMDSv2
Required
Instance ARN
arn:aws:ec2:eu-north-1:034222085276:instance/i-0c8b73d6dcfe81c16
Managed
false
Operator
-
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search
[Alt+S]
Europe (Stockholm)
Manishaperugu_21 (034222085276)
Manishaperugu_21
EC2 / Instances / i-0c8b73d6dcfe81c16
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance details Info
AMI ID
ami-04005916a1c241ff5
AMI name
al2023-ami-2023.12.20261109.0-kernel-6.18-x86_64
Stop protection
Disabled
Instance reboot migration
Default (On)
Stop-hibernate behavior
Disabled
State transition reason
-
State transition message
-
Owner
034222085276
Current instance boot mode
uefi
Answer RBN DNS hostname IPv4
Enabled
Monitoring
disabled
Allowed image
-
Launch time
Thu Nov 12 2026 10:39:28 GMT+0530 (India Standard Time)
(3 minutes)
Instance auto-recovery
Default
AMI Launch index
0
Credit specification
unlimited
Usage operation
RunInstances
Enclaves Support
Disabled
Allow tags in instance metadata
Disabled
Platform details
Linux/UNIX
Termination protection
Disabled
AMI location
amazon/al2023-ami-2023.12.20261109.0-kernel-6.18-x86_64
Lifecycle
normal
Key pair assigned at launch
Manisha@54
Kernel ID
-
RAM disk ID
-
Boot mode
uefi-preferred
Use RBN as guest OS hostname
Disabled
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search
[Alt+S]
Europe (Stockholm)
Manishaperugu_21 (034222085276)
Manishaperugu_21
EC2 / Instances / i-0c8b73d6dcfe81c16 / Connect to instance
Connect to Linux instance Info
Connect to an instance to test, troubleshoot, deploy code, update software etc. You can use browser based methods or an SSH client.
Instance info
Instance
i-0c8b73d6dcfe81c16 (Manisha@54)
State: Running
Status: 3/3 checks passed
VPC
vpc-00d6b8d3a0ffdccb4
Subnet
subnet-0fe438cffb9dea14e
Security groups
sg-096202d1aec24b8d5 (launch-wizard-1)
In web browser
In SSH client
Choose how to connect
Public subnet access
Managed SSH keys
EC2 Instance Connect
Use for instances with public IPs or accessible via standard routing. Requires network connectivity to your instance. Connect using temporary SSH keys managed by AWS.
Private subnet access
Managed SSH keys
EC2 Instance Connect Endpoint
Use for instances in private subnets. Traffic is routed through a VPC endpoint. Connect using temporary SSH keys managed by AWS.
Maximum security
IAM authentication
SSM Session Manager
Uses IAM-based authentication. Connect without SSH keys, bastion hosts, or opening any inbound ports.
Direct
Troubleshoot
EC2 Serial Console
Access your instance's serial port for troubleshooting boot, network, and OS issues. Works without network connectivity. Requires account-level enablement.
Settings
Choose public IP address
IPv4 address (56.228.35.107)
IPv6 address
Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user. Read your AMI usage instructions to check if the AMI owner has changed the default AMI username.
ec2-user
Cancel
Connect assist
Connect
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences





c) S3 Bucket Creation and Upload Objects

I. Create an S3 Bucket for file storage.

II. Upload 5 objects with different file extensions.

Aim:

To create an Amazon S3 bucket for cloud-based file storage and upload five objects with different file extensions into the bucket.

Description:

Amazon Simple Storage Service (Amazon S3) is a cloud storage service provided by AWS. It allows users to store, retrieve, and manage data over the internet.

An S3 bucket is a container used to store files, which are called objects. S3 supports different file formats such as .txt, .pdf, .jpg, .png, and .docx.

In this experiment, an S3 bucket is created and five objects with different file extensions are uploaded to demonstrate cloud-based file storage.

Steps :

I. Create an S3 Bucket

1. Log in to the AWS Management Console.
2. Search for S3 in the AWS search bar.
3. Open the Amazon S3 service.
4. Click on Create bucket.
5. Enter a unique bucket name, for example, manisha-cloud-storage-2026.
6. Select the AWS Region (use the required region for your experiment).
7. Keep Object Ownership as ACLs disabled (recommended).
8. Keep Block all public access enabled for security.
9. Leave the remaining settings as default.
10. Click Create bucket.
11. Verify that the bucket is successfully created and displayed in the S3 bucket list.

II. Upload 5 Objects with Different File Extensions

1. Open the newly created S3 bucket.
2. Click on the Upload button.
3. Click Add files.
4. Select five files with different extensions from your computer.

Example files:

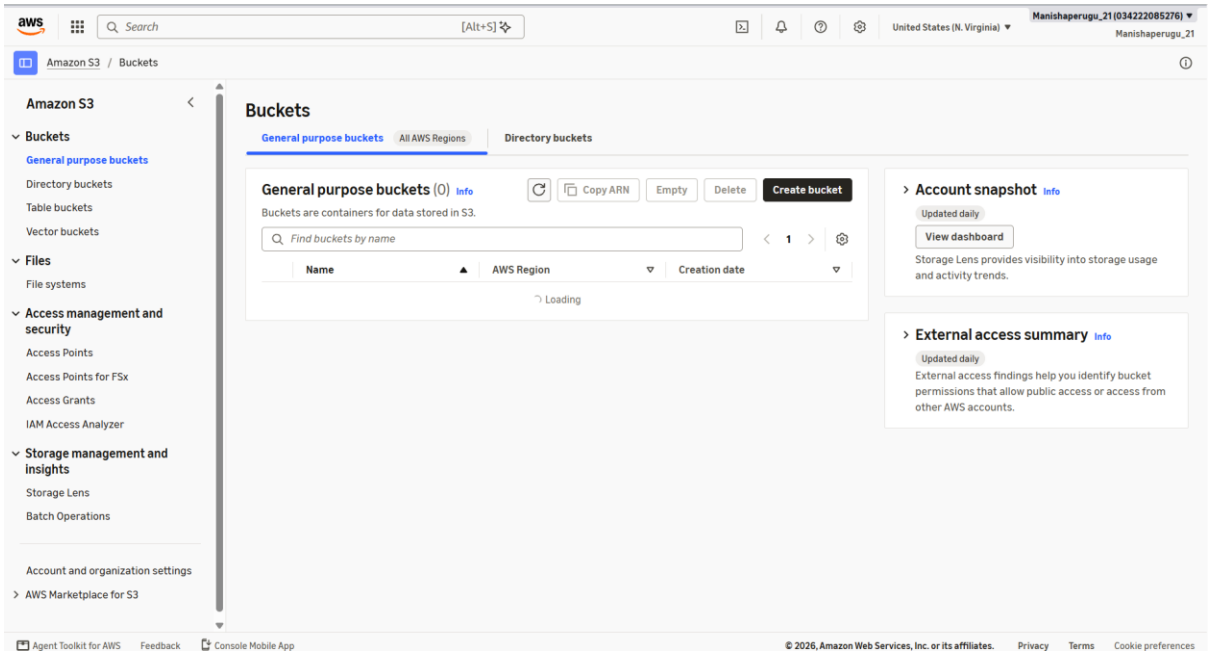
S.No.	File Name	Extension

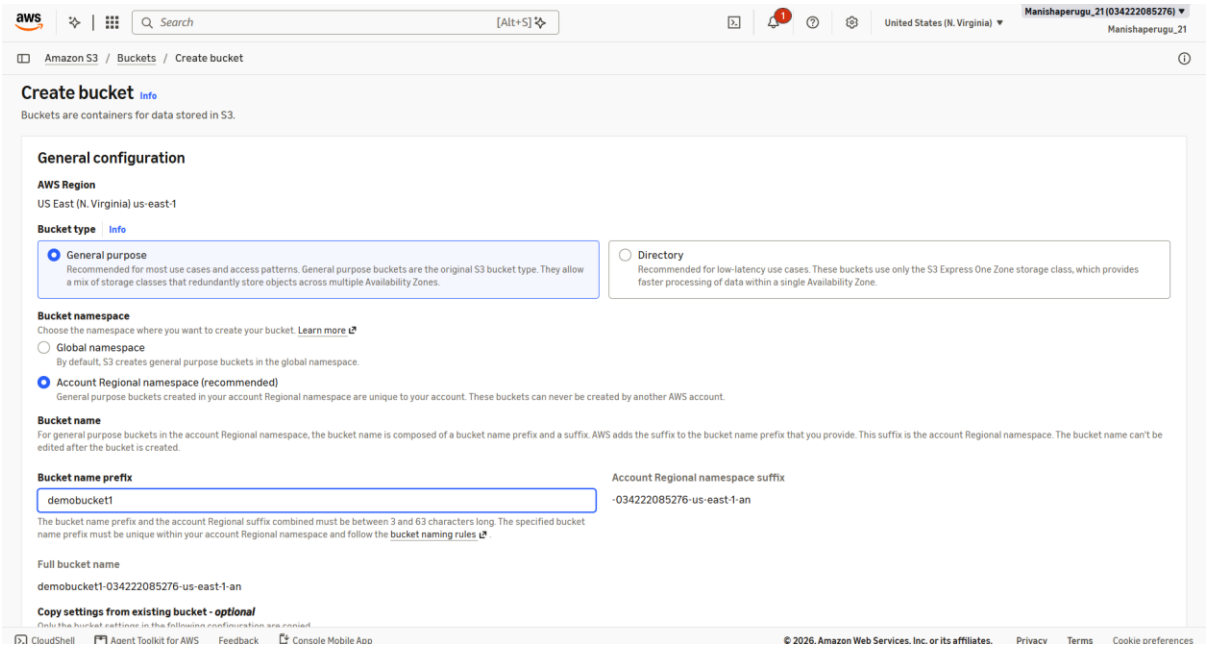
1	notes.txt	.txt
2	document.pdf	.pdf
3	image.jpg	.jpg
4	presentation.pptx	.pptx
5	data.csv	.csv

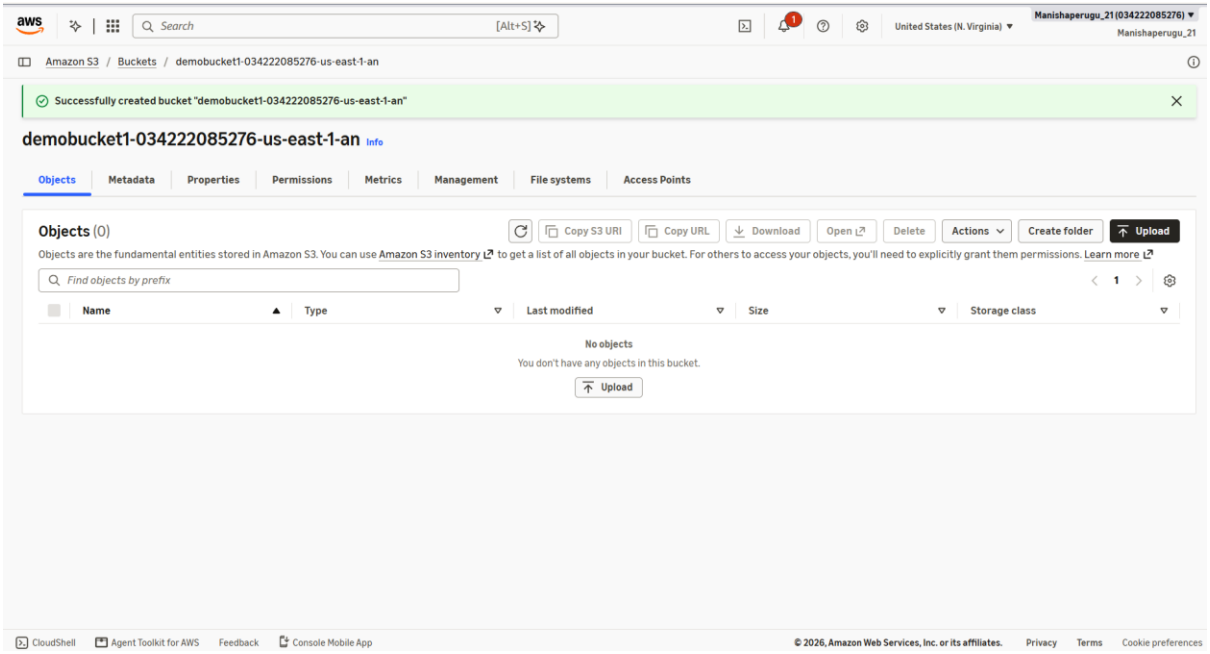
5. Verify that all five files are listed in the upload section.
6. Click Upload.
7. Wait until the upload process is completed.
8. Return to the bucket's Objects tab.
9. Verify that all five uploaded objects are displayed successfully.

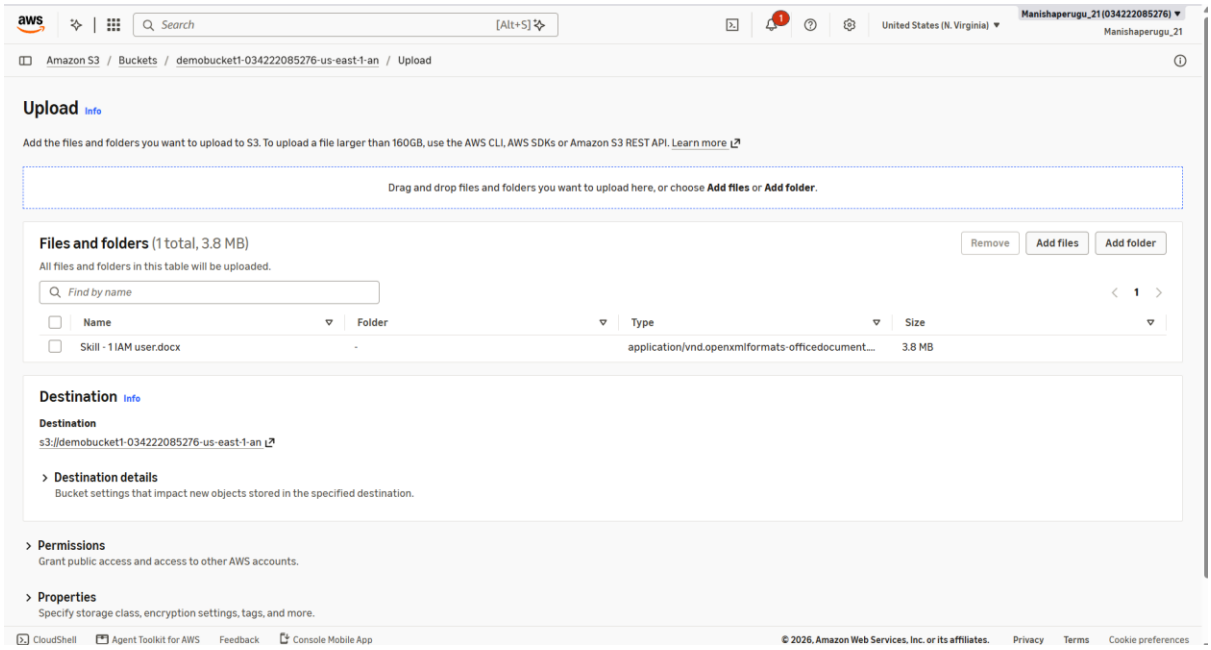
Ouput:

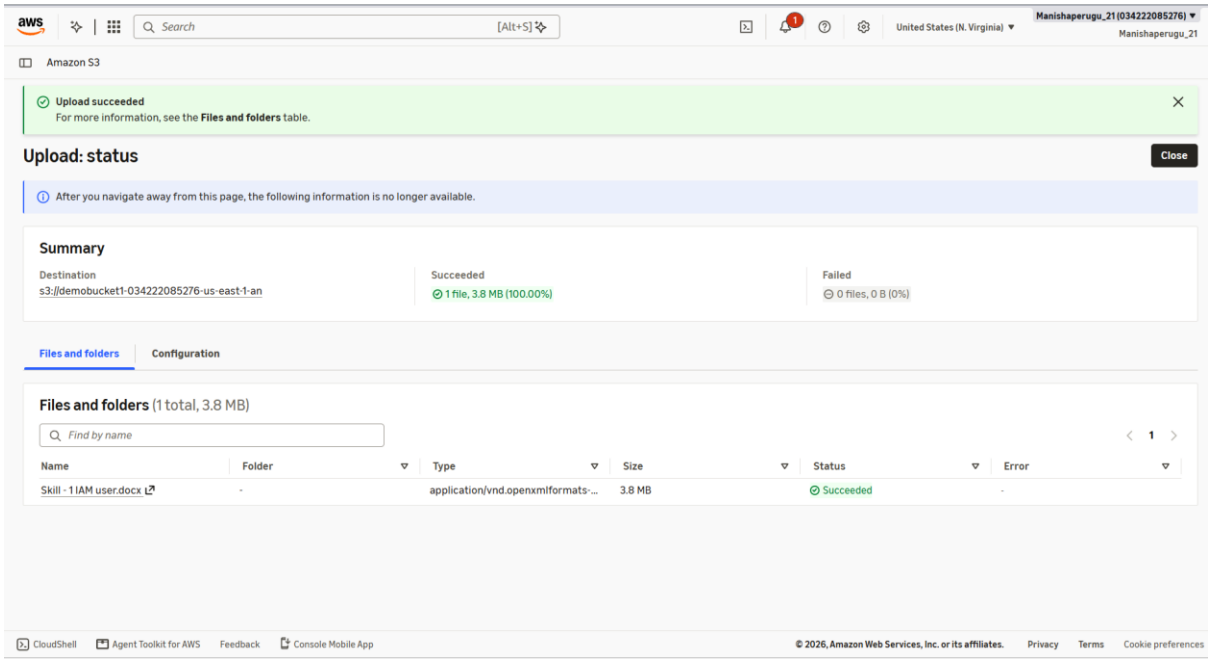
An Amazon S3 bucket was successfully created, and five objects with different file extensions (.txt, .pdf, .jpg, .pptx, and .csv) were uploaded successfully.




Search
[Alt+S]
United States (N. Virginia)
Manishaperugu_21 (034222085276)
Manishaperugu_21
Amazon S3 / Buckets / Create bucket
Create bucket Info
Buckets are containers for data stored in S3.
General configuration
AWS Region
US East (N. Virginia) us-east-1
Bucket type Info
General purpose
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
Directory
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.
Bucket namespace
Choose the namespace where you want to create your bucket. Learn more
Global namespace
By default, S3 creates general purpose buckets in the global namespace.
Account Regional namespace (recommended)
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.
Bucket name
For general purpose buckets in the account Regional namespace, the bucket name is composed of a bucket name prefix and a suffix. AWS adds the suffix to the bucket name prefix that you provide. This suffix is the account Regional namespace. The bucket name can't be edited after the bucket is created.
Bucket name prefix
demobucket1
Account Regional namespace suffix
-034222085276-us-east-1-an
The bucket name prefix and the account Regional suffix combined must be between 3 and 63 characters long. The specified bucket name prefix must be unique within your account Regional namespace and follow the bucket naming rules
Full bucket name
demobucket1-034222085276-us-east-1-an
Copy settings from existing bucket - optional
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search
[Alt+S]
United States (N. Virginia)
Manishaperugu_21 (034222085276)
Manishaperugu_21
Amazon S3 / Buckets / demobucket1-034222085276-us-east-1-an
Successfully created bucket "demobucket1-034222085276-us-east-1-an"
demobucket1-034222085276-us-east-1-an Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Access Points
Objects (0)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
No objects
You don't have any objects in this bucket.
Upload
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search
[Alt+S]
United States (N. Virginia)
Manishaperugu_21 (034222085276)
Manishaperugu_21
Amazon S3 / Buckets / demobucket1-034222085276-us-east-1-an / Upload
Upload Info
Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. Learn more
Drag and drop files and folders you want to upload here, or choose Add files or Add folder.
Files and folders (1 total, 3.8 MB)
All files and folders in this table will be uploaded.
Remove
Add files
Add folder
Find by name
Name
Folder
Type
Size
Skill - 1 IAM user.docx
application/vnd.openxmlformats-officedocument....
3.8 MB
Destination Info
Destination
s3://demobucket1-034222085276-us-east-1-an
Destination details
Bucket settings that impact new objects stored in the specified destination.
Permissions
Grant public access and access to other AWS accounts.
Properties
Specify storage class, encryption settings, tags, and more.
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search
[Alt+S]
United States (N. Virginia)
Manishaperugu_21 (034222085276)
Manishaperugu_21
Amazon S3
Upload succeeded
For more information, see the Files and folders table.
Upload: status
Close
After you navigate away from this page, the following information is no longer available.
Summary
Destination
s3://demobucket1-034222085276-us-east-1-an
Succeeded
1 file, 3.8 MB (100.00%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (1 total, 3.8 MB)
Find by name
Name
Folder
Type
Size
Status
Error
Skill - 1 IAM user.docx
application/vnd.openxmlformats-...
3.8 MB
Succeeded
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences